Pharmaceutical Organic Chemistry

Science

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Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Pharmaceutical Organic Chemistry (A07727)

Short Title: Pharmaceutical Organic Chem
Faculty Science and Computing
Department Science
Cluster Chemistry
Author CLENNON
Credits 5 **Level:** Introductory

Description of Module / Aims

This module will introduce reactions and mechanisms of key organic functional groups. Examples to illustrate the course will be drawn from biologically and pharmaceutically important compounds. The laboratory course will include a range of organic syntheses to complement the theory section. Products will be characterised via a range of analytical techniques.

Programmes

CHEM-0009	BSc (Hons) in Pharmaceutical Science (WD_SPHSC_B)	2 / 4 / M
CHEM-0009	BSc in Pharmaceutical Science (WD_SPHSC_D)	2 / 4 / M

Indicative Content

- Functional group interconversions to include carbonyls, alkenes, amines, alcohols, ethers and epoxides
- Aromatic compounds, electrophilic substitution, effect of substituents on reactivity; aromaticity, basicity and reactivity of nitrogen heterocycles
- Carbon-carbon bond formation using organometallic reagents
- Examples of biological and pharmaceutically important compounds to illustrate the above reactions
- Practical organic chemistry: a range of preparative work designed to teach the organic techniques and illustrate reactions

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Explain the mechanism of a number of fundamental organic reactions and account for the trends in reactivity.
2. Predict and explain the influence of substrate substituents on these reactions.
3. Identify synthetic routes for some reaction pathways, including functional group transformations and some carbon-carbon bond formations.
4. Employ a range of synthetic organic reactions in the laboratory, including some multi-step synthetic procedures.
5. Practise purification and characterisation techniques on synthetic products according to given procedures..

Learning and Teaching Methods

- Lectures.
- Laboratory-based practicals.
- Tutorials.
- Self-directed study.

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	50%	1,2,3
Continuous Assessment	50%	1,2,3,4,5
Tutorial/Problem Sheets	10%	1,2,3
Lab Report	40%	4,5

Assessment Criteria

<40%: No understanding of the basic principles of pharmaceutical organic chemistry. Inability to carry out associated laboratory work to a satisfactory standard.

40%-49%: Will be able to show a basic knowledge of pharmaceutical organic chemistry and demonstrate the ability to carry out associated practical work to a satisfactory standard.

50%-59%: As well as the above, the student will be able to demonstrate an understanding of some of the complexities of the topics and predict some trends in reactivity and behaviour. In addition, the learner will show a consistently good ability as reflected in associated laboratory experiments.

60%-69%: In addition, the student will demonstrate a more detailed knowledge of the all topics and demonstrate an ability to evaluate detailed knowledge acquired in the module to new material. Will demonstrate very good ability in most laboratory experiments and preparation of laboratory reports.

70%-100%: The learner will be able to demonstrate comprehensive knowledge and understanding of pharmaceutical organic chemistry, predict general trends in behaviour, including exceptional behaviour and show the ability to evaluate and relate topics. Will demonstrate excellent laboratory skills and a consistently high ability in analysis of results.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	24	
Tutorial	12	
Practical	24	
Independent Learning	75	

Essential Material

- Bruice, P.Y. _Organic Chemistry_. 6th ed. NewYork: Pearson Education, 2010.

Supplementary Material

- Ege, S.N. _Organic Chemistry: Structure and Reactivity_. 4th ed. Boston: Houghton Mifflin, 1999.
- Hart, H., D. Hart and L. Craine. _Organic Chemistry_. 10th ed. Boston: Houghton Mifflin, 2000.
- McMurry, J.E. _Organic Chemistry_. 5th ed. London: Brooks Cole, 1999.

Requested Resources

- Room Type: Chemistry Lab
- Lecture Room: Loose Seated